# Do Now · retrieval

## Lesson 063 · Thinking Logically: Conditions and Decisions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What does a Boolean expression evaluate to?

### Q2

What is a decision boundary?

### Q3

Why is condition order important in else-if?

### Q4

Simplify (A AND B) OR (A AND NOT B), showing the Boolean laws used.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Identify — Identify decisions in the solution and write precise Boolean conditions, including valid-data boundaries.

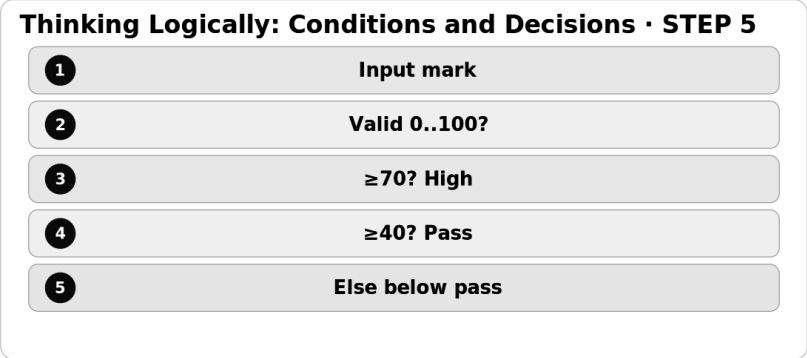
## Core idea 2

Model — Order conditions so cases are mutually understood: mark>=70 must be considered before mark>=40.

## Core idea 3

Evaluate — Use truth tables or decision tables to test normal, erroneous and boundary input values.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

predicate — Boolean condition.

selection — Branch based on a condition.

boundary test — Input near comparison threshold.

short-circuit — Skipping redundant Boolean evaluation.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

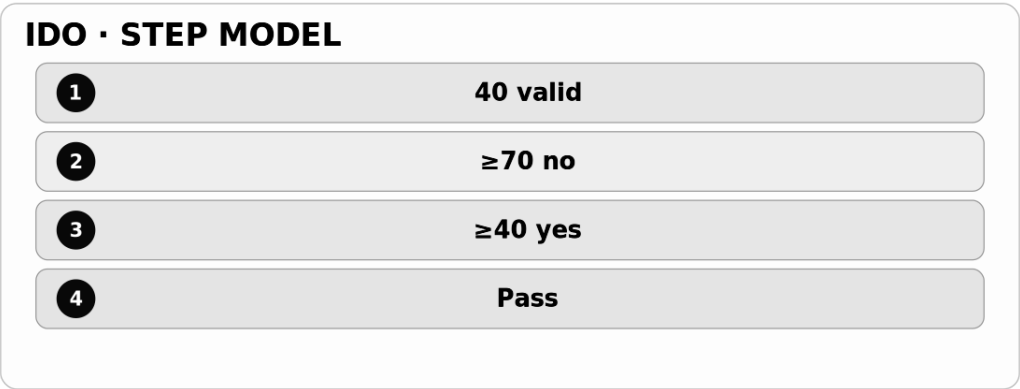
## Problem

Trace Grade(40) using the conditions shown.

## Follow the worked method

1. Explain the decision — Check comparisons in order; stop after first matching return.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Explain the decision
Without range check, 105 would incorrectly be classified High.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Trace Grade for marks 39,40,69,70 and give output at each. [4]

## Question 2

Write a Boolean rule for a valid age 11–18 inclusive, and give two boundary tests. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

A program checks if(mark>=40) before else-if(mark>=70), then outputs Pass for 85. Explain bug and repair. [6]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

With else-if, a later condition always runs even if an earlier branch matches.

## Use key vocabulary from memory

1. Explain predicate in your own words.

2. Explain selection in your own words.

3. Explain boundary test in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.